Alessio Amaolo

Ph.D. Candidate, Princeton University
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Education

Princeton University – Ph.D. Chemistry and Materials — Expected May 2027
Advised in Electrical and Computer Engineering by Alejandro W. Rodriguez.
Thesis: Physical Limits in Photonics: From Radiative Heat Transfer to Information Capacity and Design

Princeton University – M.A. Chemistry (Pass with distinction) — 2023

California Institute of Technology – B.S. Chemistry — 2021

Publications (* indicates equal contribution)

Published

1. Chao, P.*, **Amaolo, A.***, Molesky, S., Rodriguez, A.W. Bounds as blueprints: towards optimal and accelerated photonic inverse design. *Optics Express*. 2026.
2. **Amaolo, A.***, Chao, P.*, Strekha, B., Clarke, S., Mohajan, J., Chen, F.J., Molesky, S., Rodriguez, A.W. Maximum Shannon capacity of photonic structures. *npj Nanophotonics*. 2026.
3. Strekha, B., **Amaolo, A.**, Mohajan, J., Chao, P., Molesky, S., Rodriguez, A.W. Limitations on bandwidth-integrated passive cloaking. *Physical Review A*. 2024.
4. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. Physical limits on Raman scattering: the critical role of pump and signal co-design. *Physical Review A* (Letter). 2024.
5. Maldonado, T.J., Pham, D.N., **Amaolo, A.**, Rodriguez, A.W., Türeci, H.E. Negative electrohydrostatic pressure between superconducting bodies. *Physical Review B*. 2024.
6. **Amaolo, A.**, Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. Can photonic heterostructures provably outperform single-material geometries? *Nanophotonics*. 2024.
7. McNicholas, B.J., Nie, C., Jose, A., Oyala, P.H., Takase, M.K., Henling, L.M., Barth, A.T., **Amaolo, A.**, Hadt, R.G., Solomon, E.I., Winkler, J.R., Gray, H.B., Despagnet-Ayoub, E. Boronated Cyanometallates. *Inorganic Chemistry*. 2022.

Preprints / Under Review

1. **Amaolo, A.**, Molesky, S., Rodriguez, A.W., Shaker, G., Johnson, S.G., Chao, P. Gearbox cavities: ultrahigh wave confinement via broken rotational symmetry. *arXiv:2610.42003*. 2026.
2. Chen, F.J., **Amaolo, A.**, Chao, P., Molesky, S., Lin, Z., Rodriguez, A.W. Optically Incoherent Photonic Mutual Information. *arXiv:2607.13153*. 2026.
3. Virally, P., Chao, P., **Amaolo, A.**, Rodriguez, A.W., Molesky, S. How many channels can a photonic system support? *arXiv:2510.01128*. 2025.
4. Molesky, S., Chao, P., **Amaolo, A.**, Rodriguez, A.W. Inferring structure via duality for photonic inverse design. *arXiv:2504.14083*. 2025.

In Preparation

1. **Amaolo, A.**, Chen, F.J., Chao, P., Maldonado, T.J., Strekha, B., Molesky, S., Rodriguez, A.W. Optimal inverse design in nanophotonics: a tutorial. *Advances in Optics and Photonics (Invited review paper)*. In preparation.
2. Meiburg, A., *et al.* (including **Amaolo, A.**). BoundBeatingBench: A Lean-Formalized Dataset for Improving Theoretical Bounds across Scientific Domains. In preparation.

Research Experience

Student Researcher, Google DeepMind — Summer 2025
Research on computational lithography and mask process correction for chip fabrication optimization.

Graduate Researcher, Rodriguez Group, Princeton University 2021 – Present
Computing fundamental limits on photonic device performance via convex optimization and Lagrangian duality, and using these limits to guide inverse design; applications include nonlinear optics, cloaking, radiative heat transfer, and the information capacity of photonic systems.

Summer Undergraduate Research Fellow, Caltech 2018 – 2020

- Chan Group (2020): tensor networks for simulating many-body quantum systems.
- Gray Group (2019): ligand field theory and electronic structure of boronated cyanometallates via spectroscopy.
- Gray Group (2018): rhenium-based catalysts for electrochemical CO_2 reduction, with the Jet Propulsion Laboratory.

Software

dolphindes – Library for computing performance bounds on photonic devices; core contributor and maintainer.
fluxmax – Distributed JAX library for designing photonic metasurfaces for near-field radiative heat transfer.

Conference Talks

Metapix Annual Workshop (Chalmers University of Technology), Gothenburg, Sweden September 2026
"Physical Limits in Photonics: From Radiative Heat Transfer to Information Capacity and Design" – Keynote

20th International Congress on Artificial Materials for Novel Wave Phenomena, New York, USA August 2026
"Communicating through waves: How many optical channels can a photonic device support?" – Invited Talk

16th International Conference on Metamaterials, Photonic Crystals and Plasmonics, Dublin, Ireland July 2026
"The photonic gearbox: ultra-high local density of states via cylindrical-symmetry breaking" – Invited Talk

Photonics North, Quebec City, Canada June 2026
"Certificates of Suboptimality & Opportunities for New Photonic Designs" – Invited Talk

Teaching

Teaching Assistant, **Electronic Circuit Design, Analysis and Implementation** – reviews Spring 2026
Teaching Assistant, **Robot Planning Meets Machine Learning** Fall 2025
Teaching Assistant, **Optical and Quantum Electronics** – reviews Fall 2024
Teaching Assistant, **Electromagnetism for Engineers** – reviews Spring 2024, Spring 2025
Teaching Assistant, **Ecology and Evolutionary Biology** – reviews Fall 2023
Teaching Assistant, **Electromagnetism** – reviews Spring 2023
Teaching Assistant, **General Chemistry** – reviews Fall 2022
Teaching Assistant, **Statistical Mechanics** – reviews Spring 2021
Teaching Assistant, **Physical Chemistry** – course notes Fall 2019

Selected Student Teaching Feedback

- *Electromagnetism for Engineers:* "Alessio was extremely helpful, cared about our understanding of the material, and was a very fair and kind individual."
- *Statistical Mechanics:* "Alessio was an amazing TA. He was always very kind in the way he answered questions without making me feel like it was a dumb question. One of the best physics TAs I've encountered."
- *General Chemistry:* "I feel as if most of my understanding of the course came from Alessio and his inherent ability to teach a group of students effectively."

Mentorship

Francis Chen, Ph.D. student, Rodriguez Group 2024 – Present
Nicolas Wyszkowski, Ph.D. student, Rodriguez Group 2026 – Present
Zheming Li, Ph.D. student, Rodriguez Group 2026 – Present

Professional Service

Journal Reviewing

- *Journal of the Optical Society of America B*
- *Computational and Applied Mathematics*
- *Optica*
- *ACS Photonics*
- *Science Advances*

Awards

Princeton Chemistry Department David V. Milligan '62 Fellowship 2021 – 2022

- Awarded by faculty in recognition of academic excellence

Caltech Summer Undergraduate Research Fellowships: Reed and Ruth Brantley (2020), Donald Voet and Jerome Vinograd (2019), Dr. and Mrs. Daniel C. Harris (2018) 2018 – 2020